A Formative Methodology To Elicit Feedback Preferences for Partial Automation in Video Games

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Abstract

Partial automation improves video game accessibility for players with upper-limb impairments by delegating control of inaccessible game actions to software agents. A critical requirement of these systems is communicating to the player which actions the software agent performs. However, no prior work has systematically investigated which feedback modalities are most appropriate for communicating different actions. To address this gap, this paper proposes a formative elicitation methodology that adapts established elicitation approaches to this specific context. A preliminary study was conducted with three participants with upper-limb impairments playing The Ultimate Doom. Its objective was to assess the feasibility of the proposed methodology for investigating players' feedback preferences. Initial results suggest that the methodology can successfully elicit feedback preferences and stimulate participants to reason about when feedback is needed and how it should be designed. This work lays the foundation for a larger-scale study and provides initial evidence that gesture elicitation methodologies can be successfully adapted to elicit feedback preferences.

CCS Concepts

• **Human-centered computing** → **Accessibility design and evaluation methods**; *Empirical studies in accessibility*; *User centered design*; • **Applied computing** → Computer games.

Keywords

Accessibility; Video Games; Partial Automation; Feedback Elicitation; Gesture Elicitation

ACM Reference Format:
Matteo Manzoni. 2026. A Formative Methodology To Elicit Feedback Preferences for Partial Automation in Video Games. In *The 28th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '26), October 25–28, 2026, Vila Nova de Gaia, Portugal.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3797867.3845309

1 Introduction

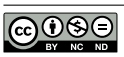
Upper-limb impairments can limit a person's ability to use standard video game controllers, creating accessibility barriers that require solutions such as input remapping [27, 31], adaptive controllers [17, 21, 36], or alternative interaction modalities [4, 34, 38, 44]. When these approaches are insufficient, partial automation enables players to delegate specific game actions to a software agent [2, 14]. Partial automation has been used in commercial games for specific tasks, such as aim assistance in the *Call of Duty* video game series [1], and in research to enable players with disabilities to play entire games autonomously [3, 13–15, 35].

Beyond gaming, partial automation has been studied across diverse domains, including collaborative robot control [8, 20, 24, 32] and assisted vehicle driving [7, 28]. In these domains, a critical requirement of partial automation systems is feedback [29, 39]. Specifically, the agent executing the automated actions must provide users with feedback that allows them to understand what the system is doing, thereby supporting coordinated control of the remaining actions. However, most partial automation systems for video games provide no explicit feedback about automated actions [3, 13, 14, 35], potentially leading to coordination difficulties [2, 3] and *automation confusion* [15] during gameplay.

Designing appropriate feedback for partial automation gaming systems is particularly challenging [16, 29, 41]. When partial automation is used as an accessibility solution, feedback preferences may also vary substantially across users due to differences in individual needs, abilities, gaming habits, and available hardware. As a result, players may wish to customize how automated actions are communicated during gameplay. However, defining personalized feedback can be difficult for users, because video game feedback spans a large design space, including visual, auditory, and haptic modalities that can be implemented in many different ways [6, 11, 19, 23]. Instead of allowing players to define the feedback from scratch, partial automation systems should therefore provide suitable default feedback configurations that users can subsequently adapt to their preferences. Nevertheless, no prior work has investigated which feedback modalities users prefer for different types of automated actions in video games.

To address this gap, this paper presents a formative elicitation methodology that adapts established gesture elicitation approaches [42, 45]. The proposed methodology investigates feedback preferences by presenting participants with automated actions during partial automation gameplay and asking them to indicate their preferred feedback modalities (*e.g.*, visual, haptic, or auditory). Aggregating participants' proposals then yields the final set of default feedback modalities.

To assess the feasibility of the proposed approach, this paper presents a preliminary case study with three players with upper-limb impairments playing *The Ultimate Doom* in partial automation. Given the limited sample size, the results are analyzed in a purely descriptive manner. The findings provide initial insights into factors



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ASSETS '26, Vila Nova de Gaia, Portugal
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ACM ISBN 979-8-4007-2521-0/2026/10
https://doi.org/10.1145/3797867.3845309

that may influence feedback preferences, including action characteristics, hardware constraints, and gaming conventions. More importantly, they suggest that the proposed elicitation protocol is feasible and can successfully capture participants' preferences and reasoning about feedback design. These results support conducting a larger-scale study to further evaluate the methodology.

2 Related Work

This section first reviews feedback in video games and its role in partial automation systems (Section 2.1). It then examines existing elicitation approaches and motivates the adoption of Wobbrock et al.'s gesture elicitation methodology as the foundation of this work (Section 2.2).

2.1 Feedback for Partial Automation Gaming Systems

In partial automation systems, feedback is used to communicate the actions performed by the software agent to the user, supporting effective human-agent coordination [16, 29, 39]. In video games, such feedback can be delivered through visual, auditory, and haptic modalities [5, 9, 37]. Each modality offers different strengths and limitations and can be tailored to accommodate user needs and preferences.

Visual feedback is conveyed through on-screen graphical elements, such as prompts providing direct instructions or icons indicating points of interest and object states. Key parameters include rendering characteristics (*e.g.*, color and contrast), movement (dynamic vs. static elements), size, and spatial placement [6, 19].

Auditory feedback can be delivered through external speakers, headphones, or controller-integrated speakers and is generally divided into verbal and non-verbal categories. Non-verbal feedback includes auditory icons, earcons, and musicons, while verbal feedback relies on natural or synthetic speech [23]. In both cases, feedback can also vary in volume and spatialization [12].

Haptic feedback in video games primarily relies on vibrotactile feedback delivered through the controller. Force feedback (providing resistance to player input) has also been introduced but remains limited to specific hardware platforms such as the PlayStation 5 DualSense controller [40]. Consequently, this work focuses exclusively on vibrotactile feedback. Typical design parameters include intensity, duration, and rhythm, defined as temporal patterns of pulses and pauses [11].

Together, these modalities offer numerous design parameters [6, 11, 19, 23]. Consequently, determining how a game action should be communicated to players in a partial automation system is not straightforward, as the same action can be associated with many alternative feedback solutions. Additionally, research on feedback for partial automation gaming systems is limited. Cimolino et al. [15] showed that the absence of explicit feedback during partial automation gameplay can lead to confusion about the division of control between player and software agent. However, no prior work has systematically investigated which feedback modalities players prefer for communicating different automated actions in partial automation gaming contexts.

2.2 Elicitation Methodologies

To study user preferences, different elicitation methodologies can be used. Several methods ask participants to evaluate and compare a given set of options – for example, best–worst scaling [33], Likert scales [30], pairwise comparisons [10], and rank ordering [18]. By aggregating multiple evaluations, these methods reveal which options are generally favored and how they relate to each other in terms of preference. However, all these methodologies require predetermining the choice set and do not provide a mechanism to resolve cases where the same option is selected as best for multiple items.

Both of these limitations are especially relevant for feedback design. Because feedback spans a large design space (Section 2.1), defining a representative set of feedback alternatives that can be proposed to the user may be challenging. Consequently, participants may be unable to express preferences for solutions that are not included in the predefined choice set. Furthermore, feedback assigned to different game actions should remain distinguishable during gameplay; otherwise, conflicts may arise when the same feedback modality is associated with multiple actions [41].

Wobbrock et al.'s gesture elicitation methodology [42, 45] addresses both limitations. In this methodology, participants are shown a system event (the "referent" [43]; for example, zoom in on a map) and asked to propose the gesture that would produce that result. The process involves three main steps: (1) participants propose gestures in response to demonstrated referents, (2) proposals are grouped by similarity, and (3) agreement is calculated to identify consensus patterns. Although originally developed for gesture design, the methodology is well suited to feedback elicitation. In both cases, the goal is to map a set of referents to user-defined artifacts (*e.g.*, gestures or feedback modalities) while exploring a large design space without constraining participants to predefined alternatives. The methodology also provides mechanisms for analyzing agreement and resolving conflicts between referents – both key requirements in the design of feedback for automated game actions. For these reasons, the methodology used in this work builds upon Wobbrock et al.'s approach.

3 Feedback Elicitation Methodology

This work proposes a feedback elicitation methodology to identify users' preferred feedback modalities for communicating the game actions performed by a software agent in a partially automated video game. The methodology focuses on a set of game actions for which feedback preferences are elicited. Following the terminology commonly adopted in elicitation studies, these actions are referred to as *referents* [42, 43].

The protocol consists of five phases, which are summarized in Fig. 1. The first four phases are conducted individually with each participant, while the fifth phase is performed by the researchers after data collection has been completed for all participants.

3.1 Phase 1: Familiarization With Partial Automation and the Game Stimulus

After a brief introduction to the study goals and structure, the supervisor collects consent and demographic data. As partial automation gaming systems are not yet widespread [2], participants cannot be

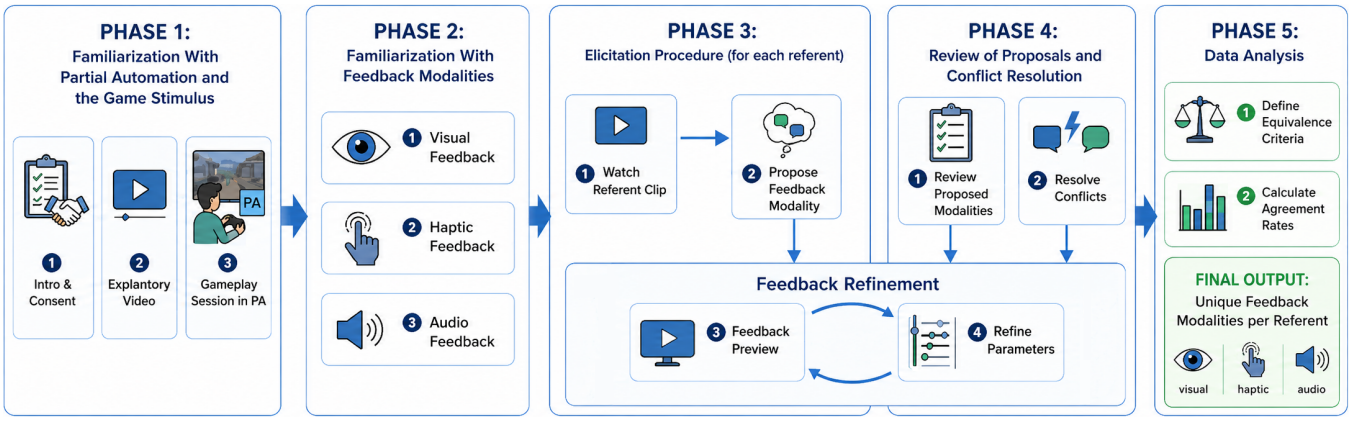


Figure 1: Overview of the proposed methodology.

expected to have prior experience with partial automation. Consequently, the study begins with a familiarization phase aimed at providing participants with a common understanding of partial automation gameplay and direct experience with the study game stimulus. First, participants watch a brief video explaining how partial automation works, introducing the game used in the study, and describing which of its actions can be automated. Then, they play a brief session of the game in partial automation, continuing until they feel confident in their understanding of partial automation gameplay. Participants are free to choose which game actions to automate.

3.2 Phase 2: Familiarization With Feedback Modalities

After the gameplay session, participants watch a second video. The video first explains that the study aims to understand when feedback is needed for automated actions and which feedback modalities are most appropriate in different situations. It then introduces participants to feedback modalities commonly used in video games (Section 2.1) by presenting examples for each modality. The goal of this phase is to ensure that participants share a common understanding of the feedback design space before proposing solutions during the elicitation phase.

3.3 Phase 3: Elicitation Procedure

For each referent, participants first watch a short gameplay clip showing the automated action being performed without any accompanying feedback. They are then asked which feedback modality they consider most appropriate for the referent. The elicitation procedure is designed to support informed feedback proposals through an iterative design process. Participants can respond freely and at any level of detail. The supervisor then helps them further specify the proposed feedback through follow-up questions, such as its appearance, timing, duration, or placement within the game. For example, for the "jump" referent, a participant may first propose a brief vibration; the supervisor would then help define its intensity and when it should be delivered. Additionally, participants can preview how the feedback would manifest in the game. To support this, a custom tool was developed that lets the supervisor set participant-specified parameters and preview the resulting feedback in real time, enabling evaluation based on actual perception rather than imagination alone. Participants are encouraged to articulate their reasoning, revisit choices, and modify parameters as needed. Participants can also declare that no feedback is necessary. This process is repeated for all referents, with referent order counterbalanced via a Latin Square design.

3.4 Phase 4: Review of Proposals and Conflict Resolution

After all referents have been elicited, participants are asked to collectively review and confirm their feedback proposals. Since participants only gain a complete view of the elicited feedback set after completing the elicitation process for all referents, they may wish to revise feedback modalities proposed earlier. Accordingly, they are given the opportunity to review and refine their proposals. Similarly, while a feedback modality may be appropriate for a specific referent in isolation, assigning similar modalities to multiple referents may reduce distinguishability during gameplay [41]. Therefore, in this phase, participants are also asked to identify and resolve potential ambiguities between the feedback assigned to different referents. As in the elicitation phase, participants can preview the revised feedback configurations before confirming their final proposals.

3.5 Phase 5: Data Analysis

The study sessions were audio-recorded and subsequently transcribed. he feedback modalities selected by participants, together with the annotations collected during the study, were then analyzed following the methodology defined by Vatavu and Wobbrock [42]. The primary challenge in applying this analysis to feedback is determining when two modalities should be considered equivalent. As in gesture elicitation, equivalence can be established through expert judgment, codebooks, or computational similarity measures [42]. Regardless of the approach, two feedback modalities should be considered equivalent if they are similar enough that players are likely to interpret them in the same way during gameplay.

Once equivalence is established, agreement rates are calculated as described by Vatavu and Wobbrock [42]. These rates quantify the extent to which participants independently proposed equivalent feedback modalities for a given referent. The resulting agreement values are then used to identify the feedback modalities with the strongest participant consensus and derive a final consensus set across all referents [45].

4 Preliminary Study

This preliminary study applies the methodology proposed in Section 3 to *The Ultimate Doom* video game. Rather than identifying consensus patterns among participants, the study aims to assess the feasibility of the methodology, exploring participants' feedback preferences and the rationale underlying them.

The study protocol was approved by the Ethics Committee of the University of Milan[1] (opinion no. 48/25). Before providing written informed consent, participants were informed of the study's goals and procedures, the risks and benefits of participation, and their right to withdraw at any time without consequence. Each study session lasted approximately one hour.

4.1 Game Stimulus

The study used *The Ultimate Doom* [25] as a stimulus (Fig. 2), a first-person shooter where players control a soldier navigating a maze-like environment in search of the exit. To progress, players must explore the map, avoid or defeat enemies, and collect resources such as weapons, ammunition, and health kits. Players can perform seven actions, which served as referents in the elicitation study (Table 1): movement, jumping, running, aiming, shooting, weapon switching, and interaction. Partial automation was integrated into the game using the *GamePals* framework [3]. The game was selected because it has a clear objective, allowing participants to quickly understand the game within the limited time frame of the study.



Figure 2: Screenshot from *The Ultimate Doom*.

4.2 Participants

Three participants (Table 2) were recruited through local associations for people with motor impairments in Milan, Italy. The only inclusion criterion was having an upper-limb impairment that limited the use of standard video game controllers. Participants with multiple disabilities were excluded to reduce potential confounding factors. Given the small sample size and the potentially identifying nature of the demographic and disability information reported in Table 2, participant data were pseudonymized using participant IDs, and age was reported as a range rather than an exact value to reduce the risk of re-identification. Experience with the game stimulus or with partial automation was not required, as the protocol included an initial familiarization phase (Section 3.1).

The group was intentionally heterogeneous, differing in impairment type, gaming habits, and preferred gaming platforms. This diversity allowed for capturing a wide range of interaction needs and design perspectives, making it suitable for a preliminary study aimed at assessing the feasibility of the proposed methodology rather than obtaining generalizable results.

[1] Planning, conduct, and reporting of the work are consistent with the general principles enunciated in [22].

4.3 Data Analysis and Taxonomy

Due to the small sample of participants, no agreement analysis or consensus-set derivation was performed (Section 3.5). Instead, the present analysis is preliminary and purely descriptive, reporting for each referent the feedback modalities selected by participants, their rationales, potential commonalities across responses, and conflicts that emerged when multiple referents shared the same feedback modality.

To determine equivalence between feedback proposals, a codebook model was adopted [42], defining a taxonomy describing the main characteristics of the elicited solutions. The taxonomy (Table 3) focuses on four dimensions that were considered relevant for distinguishing participant proposals: feedback modality, persistence, timing, and location. Two proposals were considered equivalent if they shared the same characteristics across all taxonomy dimensions. The design of the taxonomy was based on prior work on feedback design [6, 11, 12, 19, 23].

4.4 Results

Table 4 summarizes participants' proposals. Visual modalities were the most frequent (9 proposals, including the visual component of **P3**'s multimodal proposal), followed by audio (8 proposals, including the audio component of the same multimodal proposal), no feedback (3 proposals), and haptic (2 proposals). Only **P3** combined two modalities simultaneously (text + audio for the running referent).

4.4.1 Equivalent Feedback Proposals. Only one pair of proposals was classified as equivalent according to the taxonomy (Section 4.3). For the aiming referent, **P1** and **P2** independently proposed different crosshair icons displayed on the targeted enemy. Although the visual appearance of the icons differed, the proposals shared the same characteristics across all taxonomy dimensions and were therefore classified as equivalent.

Other proposals exhibited partial similarities without being considered equivalent. For both weapon switching and movement, **P1** and **P2** proposed similar feedback solutions that differed only in the timing dimension. For weapon switching, both selected a weapon icon, while for movement both proposed directional arrows

Table 1: Referents used in the preliminary study.

Game Action	Description
Movement	Moving the player character through the game environment.
Aiming	Adjusting the player's view horizontally and vertically to orient the camera and target objects or enemies.
Jumping	Performing a jump to overcome obstacles or reach elevated areas.
Shooting	Firing the currently equipped weapon.
Weapon Switching	Changing the currently equipped weapon.
Running	Increasing movement speed to traverse the environment more quickly.
Interaction	Activating or manipulating in-game objects, such as opening doors, pressing switches, or collecting items.

Table 2: Participants' demographic, disability, and gameplay characteristics.

ID	Age Range	Gender	Disability		Gameplay		
			Type	Onset Age	Freq.	Difficulty	Platform
P1	19–28	F	Merosin deficiency	Birth	Daily	Moderate	Smartphone
P2	19–28	M	Duchenne muscular dystrophy	Birth	Weekly	Low	PC
P3	39–50	M	Right hemiparesis	1 month	Weekly	Moderate	Switch

Table 3: Taxonomy of feedback.

Dimension	Values	Description
Modality	Visual (Icon / Text)	Feedback delivered visually through graphical elements.
	Auditory (Sound / Verbal)	Feedback delivered through sounds or speech.
	Haptic (Vibration)	Feedback delivered through a vibration of the controller.
	Multimodal	Combination of multiple modality–type pairs.
	None	No feedback.
Persistence	Fixed	Feedback is presented for a predefined duration.
	Dynamic	Feedback remains active while the action is being executed.
Timing	Anticipatory	Feedback starts before the action is executed.
	At onset	Feedback starts when the action begins.
Location	Static	Feedback is not associated with a specific location, direction, or entity.
	Spatialized	Feedback is associated with a specific location, direction, or entity.

indicating the movement direction. In both cases, **P1** preferred anticipatory feedback, while **P2** preferred feedback starting at the onset of the action. Similarly, **P1** and **P3** both selected a running sound effect to communicate the running referent. Unlike **P1**, however, **P3** complemented the audio cue with a text label.

Given the limited number of equivalent proposals, the remainder of the analysis focuses on feedback modalities and the rationales provided by participants, which emerged as the most informative aspects for understanding their design choices.

4.4.2 Visual Feedback. For visual feedback, only **P3** used text (*"Shoot"* and *"Run"* during execution), while **P1** and **P2** preferred icons. **P2** motivated this choice because they consider text to be more distracting during gameplay. For weapon switching, timing preferences emerged: **P1** wanted an icon appearing 1 second before the action (anticipatory), while **P2** preferred continuous confirmation. **P1** explained their choice of an icon by noting that it is easy to associate with the referent. **P1** also highlighted accessibility needs, requesting a white background for the icon to ensure sufficient contrast with the game environment. For aiming, **P1** and **P2** selected icons following the aimed enemy, signaling the optimal shooting moment. Similarly, for movement, both **P1** and **P2** proposed a directional arrow icon signaling the direction of the agent's movement.

4.4.3 Auditory Feedback. Participants favored auditory icons (*e.g.*, weapon firing and jump sounds). No participant used verbal feedback; however, **P1** requested a vocal-like sound for the jump referent – a brief *"hop"*-like confirmation sound typical of video

Table 4: Feedback proposals generated by participants for each referent.

Referent	P	Proposal
Movement	**P1**	Directional arrows displayed before movement, indicating the movement direction.
	P2	Directional arrows displayed during movement, indicating the movement direction.
	P3	Spatialized earcon played during movement, indicating the movement direction.
Aiming	**P1**	Crosshair icon displayed on the target enemy during aiming.
	P2	Crosshair icon displayed on the target enemy during aiming.
	P3	Spatialized earcon played during aiming, indicating the enemy's position.
Jumping	**P1**	Jump sound effect played at jump onset.
	P2	Haptic cue delivered immediately before the jump.
	P3	Earcon played during the jump.
Shooting	**P1**	Weapon-specific firing sound played at shooting onset.
	P2	No feedback.
	P3	*"Shoot"* text label displayed during shooting.
Weapon Switching	**P1**	Weapon icon displayed before weapon switching.
	P2	Weapon icon displayed during weapon switching until the desired weapon is selected.
	P3	Weapon-specific audio cue played at weapon-switch onset.
Running	**P1**	Running sound effect played during running.
	P2	No feedback.
	P3	*"Run"* text label and running sound effect presented during running.
Interaction	**P1**	Haptic feedback delivered during interaction.
	P2	Animated loading icon displayed during interaction.
	P3	No feedback.

games [26] – reflecting the influence of established gaming conventions. **P3** employed distinct sounds for each weapon during switching, enabling immediate recognition. Moreover, **P3** was the only participant to use earcons (for movement, aiming, and jumping). For aiming, they combined a spatialized earcon with enemy positional information, allowing the feedback to convey both the action being performed and the enemy's location.

4.4.4 Haptic Feedback. The low adoption of haptic feedback may be related to hardware constraints and individual setup choices. **P3** never selected it because they used the *Xbox Adaptive Controller* [36] without vibration, highlighting how hardware limitations influence available choices. Instead, **P1** chose haptic feedback for the interaction referent. **P2** used it to anticipate jumps, explaining that a physical cue could help alert players to actions they may not expect the agent to perform at that moment.

4.4.5 No Feedback. **P2** and **P3** proposed no additional feedback for some referents, although their motivations differed. For running, **P2** stated that they were not interested in being informed about the action. For shooting, instead, they argued that "the game's feedback is enough" and that additional feedback would interfere because "the game is very fast". Interestingly, **P1**'s proposal for shooting consisted of a weapon-specific firing sound, similar to the game's existing firing audio. This may indicate that, despite proposing an additional feedback cue, **P1** also considered the native game feedback appropriate for this action. **P3** stated instead that no feedback was needed for the interaction referent because "it's already visible".

4.4.6 Conflict Resolution. Participants rarely chose conflicting feedback modalities. **P3** noticed only at the end that they had selected auditory feedback for four actions but did not revise them, as the sounds were distinct and the actions would rarely occur simultaneously during gameplay. Similarly, **P2** initially placed the weapon switch and interaction icons in the same screen position, assuming the actions would not occur simultaneously. However, **P2** later moved the weapon switching icon to the bottom-right corner of the screen, where "you would expect the icon in a normal game".

5 Discussion

This section first reflects on the main differences between feedback elicitation and traditional gesture elicitation (Section 5.1). It then discusses the feasibility of the proposed methodology based on the findings of the preliminary study (Section 5.2). Finally, it examines the main limitations of the preliminary study and their implications for future work (Section 5.3).

5.1 Adapting Gesture Elicitation to Feedback Elicitation

The proposed methodology aims to understand users' preferences regarding feedback for automated actions in partially automated video games. While building on Wobbrock et al.'s methodology [42, 45], feedback elicitation differs from traditional gesture elicitation in five important respects. Table 5 summarizes these differences.

Traditional gesture elicitation focuses on user-generated gestures that trigger system events, while feedback elicitation focuses on system-generated outputs that communicate automated actions to users ①. As a result, participants are not asked to reason about

what they want to do, but rather about how the system should communicate information to them (2). Consequently, they must consider factors such as the cognitive effort required to interpret feedback and the ease with which feedback can be recognized, rather than the physical effort required to perform a gesture and the ease with which a gesture can be executed (3). They must also consider the potential for feedback to distract from gameplay, rather than the social acceptability of performing a gesture.

A further distinction concerns the range of proposals that participants can make. In gesture elicitation, participants are generally expected to provide a single gesture for each referent. In feedback elicitation, instead, participants may propose multimodal feedback solutions (4) or decide that no additional feedback is necessary (5). Both situations occurred during the preliminary study: **P2** proposed no feedback for the shooting and running referents, while **P3** proposed a multimodal feedback solution for the running referent and no feedback for the interaction referent. This suggests that feedback elicitation must account not only for preferences regarding feedback design, but also for judgments about whether feedback is needed at all and whether multiple feedback modalities should be combined.

5.2 Feasibility of the Proposed Methodology

The preliminary study provides encouraging evidence regarding the feasibility of the proposed methodology. All participants successfully completed the protocol and were able to propose feedback for each referent. No participant reported difficulties understanding the concept of partial automation, the elicitation task, or the role of feedback in communicating automated actions. Furthermore, participants did not simply select a feedback modality, but consistently reflected on whether feedback was needed at all and provided articulated rationales for their choices.

The results suggest that the combination of familiarization, iterative refinement, and real-time previews is effective in supporting the elicitation process. Through follow-up questions, participants progressively refined their proposals by specifying characteristics such as timing, duration, and presentation. Across all referents and participants, proposals underwent an average of 2.5 refinement iterations before being finalized. The number of iterations varied across referents, ranging from 1.3 on average for interaction to 3.7 for movement, suggesting that some referents required more refinement before participants converged on a final proposal. The preview tool further supported this process by allowing participants to experience the proposed feedback and modify it when necessary. This reduced the need to reason about feedback solely in the abstract and enabled participants to evaluate more concrete feedback alternatives.

The methodology also produced a diverse set of feedback proposals and associated rationales. Beyond the specific feedback modalities selected, participants consistently justified their choices with reference to factors such as gameplay conventions, hardware constraints, and action characteristics. This suggests that the methodology captures not only feedback preferences but also the reasoning underlying those preferences. Such information may prove valuable when interpreting participant choices and refining feedback designs in future studies.

Finally, although the present study was conducted using *The Ultimate Doom*, the proposed methodology is not tied to a specific game or genre. The protocol can be adapted to different partial automation gaming contexts by selecting appropriate referents and game stimuli, while leaving the overall elicitation procedure unchanged.

5.3 Limitations of the Preliminary Study

The preliminary study presented in this paper has three main limitations. First, the study involved only three participants. Consequently, the results are purely descriptive and no agreement-rate analysis or consensus-set derivation was performed.

Second, the taxonomy was intentionally designed to be minimal, focusing on a small set of dimensions that could be consistently identified across participant proposals. However, while it proved sufficient to support the analysis of the collected data, additional dimensions may be needed when analyzing larger datasets. For example, for the movement referent, one participant may propose a visual landmark indicating the destination toward which the software agent is moving, while another may propose a visual arrow indicating the direction of movement. Although the two proposals communicate different information about the automated movement, they would currently be classified as equivalent because they share the same modality, timing, persistence, and location. Therefore, the taxonomy may need to be extended to capture such semantic differences between feedback proposals.

Third, the equivalence classification process was performed by a single researcher. Previous elicitation studies recommend that proposals be classified by two or more independent annotators and that inter-rater agreement be assessed to establish the reliability of the classification process [42]. As a result, the reliability and reproducibility of the classification process should be further evaluated in future studies.

6 Conclusions and Future Work

This paper presents a formative elicitation methodology to investigate feedback preferences for partial automation in video games, addressing a gap in the literature: the lack of structured methods for identifying how automated actions should be communicated to players with upper-limb impairments. The proposed approach aims to derive user-informed feedback configurations that can serve as initial defaults during partial automation gameplay and subsequently be customized according to individual needs and preferences. To achieve this, the methodology combines direct experience with partial automation gameplay, iterative refinement of feedback proposals, and real-time previews of participant-defined feedback.

A preliminary study with three participants was conducted to assess the feasibility of the methodology. The results suggest that participants can successfully engage with the protocol, articulate feedback preferences, and provide detailed rationales for their choices. These findings support the use of the methodology in a larger-scale elicitation study.

Future work will apply the methodology to a larger sample of players with upper-limb impairments. This will enable the refinement and validation of the proposed taxonomy, as well as the application of established elicitation-study analysis techniques to

Table 5: Key differences between traditional gesture elicitation and feedback elicitation.

	Aspect	Gesture Elicitation	Feedback Elicitation
①	*Elicited artifact*	User-generated input used to trigger a system action.	System-generated output used to communicate an automated action.
②	*Primary question*	"What should the user do?"	"How should the system communicate this action?"
③	*Main design concerns*	Ease of execution, memorability, social acceptability.	Interpretability, noticeability, cognitive effort, potential distraction.
④	*Possibility of multimodality*	Usually a single gesture	Multiple feedback modalities can be combined
⑤	*Possibility of omission*	A gesture is required	Participants may decide that no feedback is needed

identify recurring feedback proposals and derive user-preferred feedback modalities for automated game actions.

Acknowledgments

Generative AI tools (Gemini) were used to generate Fig. 1.

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